

Which Type Of SOLAR CELL Is Best FOR YOUR APPLICATION



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A solar panel is made up of solar cells that convert solar energy (sunlight) to electricity. This conversion is known as photovoltaic effect. Solar energy is the best alternate energy source as it is renewable, easy to produce and saves the environment. The government of India too promotes it by providing remarkable subsidies.

Basic structure of a solar cell and its working principle

A solar cell, also known as a photovoltaic cell, works on the principle of photoelectric effect—ability of an object to emit electrons when light falls on it. To generate electricity from a solar cell, a material is required that emits electrons and is raised to a higher energy state by the absorption of light. These electrons have enough energy to drive the connected device or load. The cells are made of a semiconductor material that is used in the form of p-n junctions.

A solar cell consists of n-type (having electrons as a majority charge carrier) and p-type (having holes as a majority charge carrier) semiconductor materials. The depletion region is formed between n-type and p-type

materials. The depletion layer is made up of positive and negative ions. Electrons and holes are not present inside the depletion layer. Light travels in the form of packets of energy called photons. These photons (sunlight) are applied on the depletion region.

Photons are absorbed in depletion regions and generate electron-hole pairs. These free electrons and holes have enough energy to jump out of the depletion zone. If a wire is connected from the cathode (n-type material) to the anode (p-type material), electrons flow through the wire. The electrons have negative charge and, hence, get attracted by p-type material (positive terminal), travel through the external load (device) and create a flow of electrons (current), as shown in Fig. 1.

As shown in Fig. 2, the sunlight concentrator focuses sunlight on a small part and tracks the Sun throughout the day for higher efficiency. The solar panel generates unstable or fluctuating electricity and, thus, has to be regulated. The power-conditioning block gives regulated stable electricity that is stored in the battery or a battery bank. The battery gets damaged if a proper power-conditioning

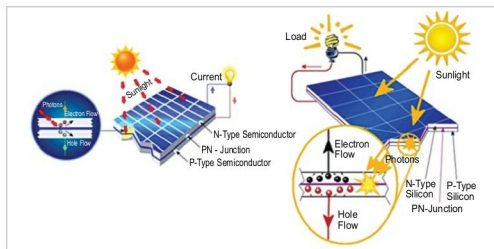


Fig. 1: Working principle of a solar cell